An Empirical Study on Configuration Robustness of Unsupervised IDS

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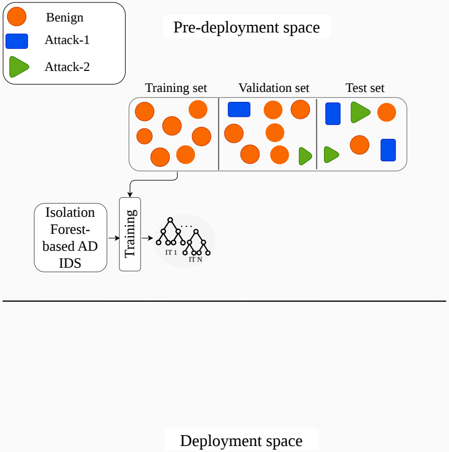
NOMS 2026

ML-based AD IDS Workflow: Train $\rightarrow$ Validate $\rightarrow$ Test $\rightarrow$ Deploy

- The model is trained on a training set $\mathcal{D}_{\textbf{train}}$.



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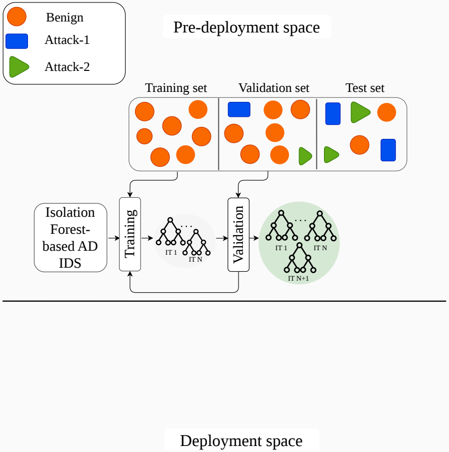

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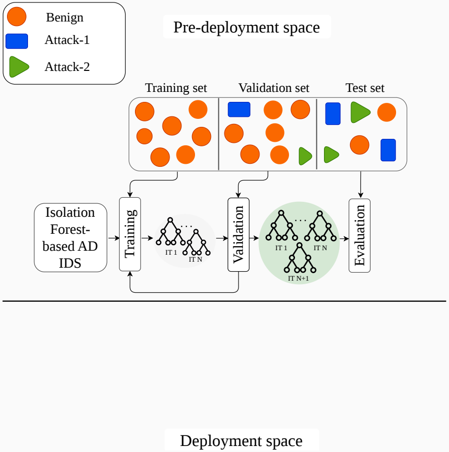

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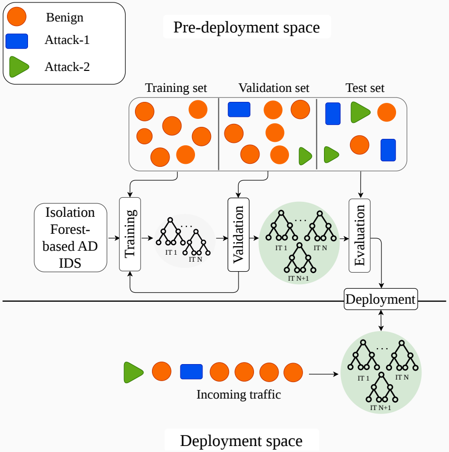

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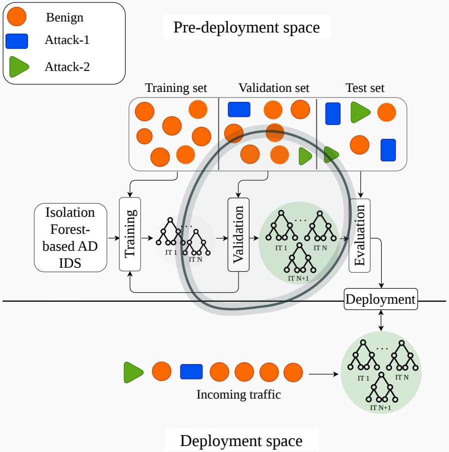


$\Longrightarrow$ Focus on the impact of the **configuration of hyperparameters (HPs)** on **AD IDS performance**.

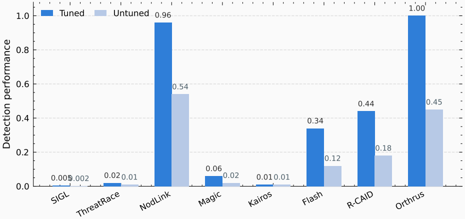
Related Work and Gap

Broader Machine Learning

- **Hutter'14 / van Rijn'18[1,2]:** a few HPs explain most performance variation.
- **DOMAINBED / WILDS / HYPERTIME[3–5]:** HPs optimized on one data distribution may fail after the distribution changes.

Security / IDS

- **Arp'22 / Pendlebury'19 / Masum'21[6–8]:** IDS results depend strongly on tuning and validation.
- **Bilot'25[9]:** 8 provenance-graph IDSs; tuned vs. untuned changes the conclusion.
- Static analyses of configuration impact.



***Source*:** **Bilot et al., USENIX Security 2025**

$\Longrightarrow$ **Gap:** no direct quantification for AD IDS under network context changes.

(i.e., traffic volume, composition, topology, and attack types)

[1] Hutter et al., "An Efficient Approach for Assessing Hyperparameter Importance", ICML 2014; [2] van Rijn and Hutter, "Hyperparameter Importance Across Datasets", KDD 2018.
[3] Gulrajani and Lopez-Paz, "In Search of Lost Domain Generalization", arXiv 2020; [4] Koh et al., "WILDS: A Benchmark of in-the-Wild Distribution Shifts", ICML 2021.
[5] Zhang et al., "HyperTime: Hyperparameter Optimization for Combating Temporal Distribution Shifts", ACM MM 2024; [6] Arp et al., "Dos and Don'ts of Machine Learning in Computer Security", USENIX Security 2022.
[7] Pendlebury et al., "TESSERACT: Eliminating Experimental Bias in Malware Classification across Space and Time", USENIX Security 2019; [8] Masum et al., "Bayesian Hyperparameter Optimization for Deep Neural Network-Based Network Intrusion Detection", IEEE BigData 2021.
[9] Bilot et al., "Sometimes Simpler is Better: A Comprehensive Analysis of State-of-the-Art Provenance-Based Intrusion Detection Systems", USENIX Security 2025.

Operational Setting and Research Questions

Operational setting

- Configuring θ can be costly; in practice, it is often fixed once.
- But the network context changes over time: traffic volume, composition, topology, and attack types evolve.

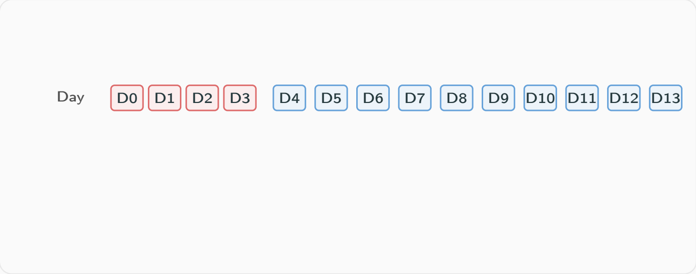
Research questions

- Can cheaper strategies approach an oracle-optimized reference without full reconfiguration?
- If θ is reused, how much performance is lost compared with reconfiguring for the current network context?

$\Longrightarrow$ Measure the performance penalty of avoiding full reconfiguration in AD IDS under changing network contexts.

Experimental Setup: Benchmark Context

- Corrected IDS2017[1] / IDS2018[2] provide day-level traffic datasets $D_0, \ldots, D_{13}$.



[1] Sharafaldin et al., "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization", ICISSP 2018.
[2] CSE-CIC-IDS2018 on AWS, CSE and CIC, 2018; official dataset page recommends citing Sharafaldin et al., ICISSP 2018.

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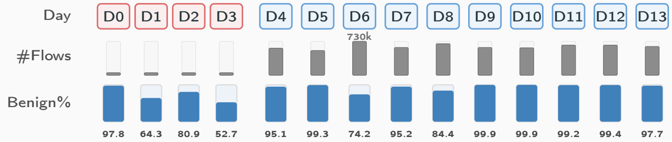

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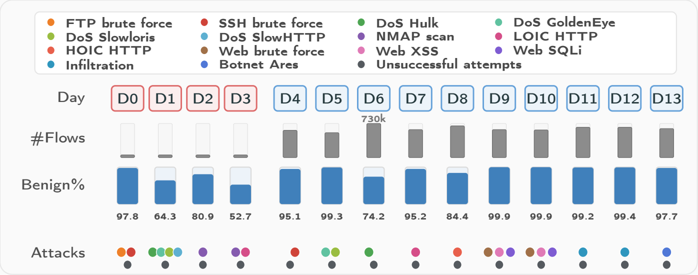

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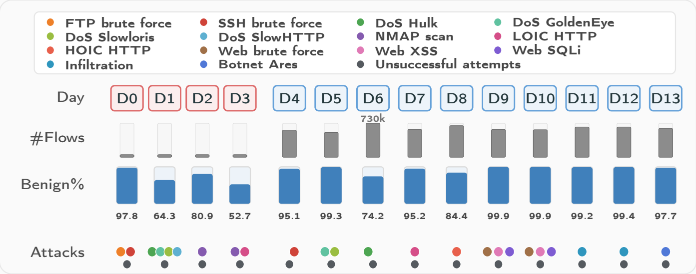

$\Longrightarrow$ Each day is considered as a specific network context; moving from one day to another simulates a context change.

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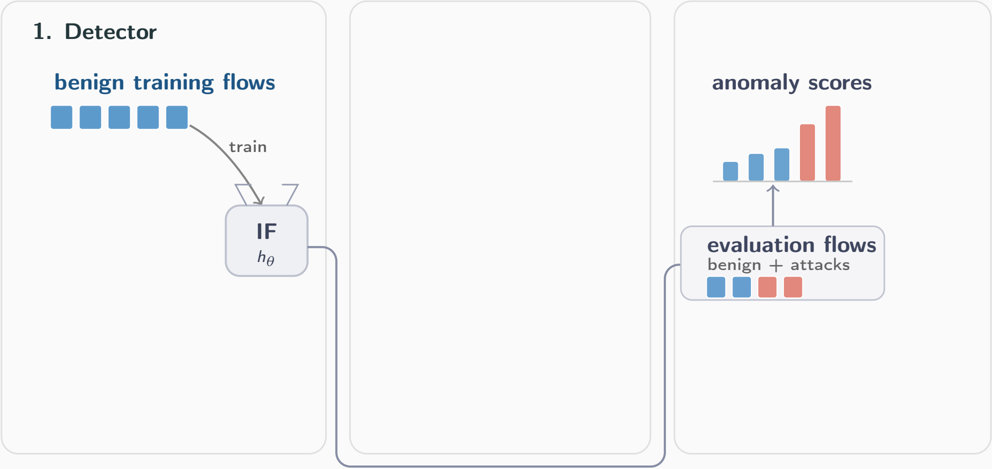
Experimental Setup: IF, BO, and Robustness Metric



$\Longrightarrow$ Isolation Forest (IF): train on benign flows, score benign + attack flows.
Lightweight enough for many configuration runs.

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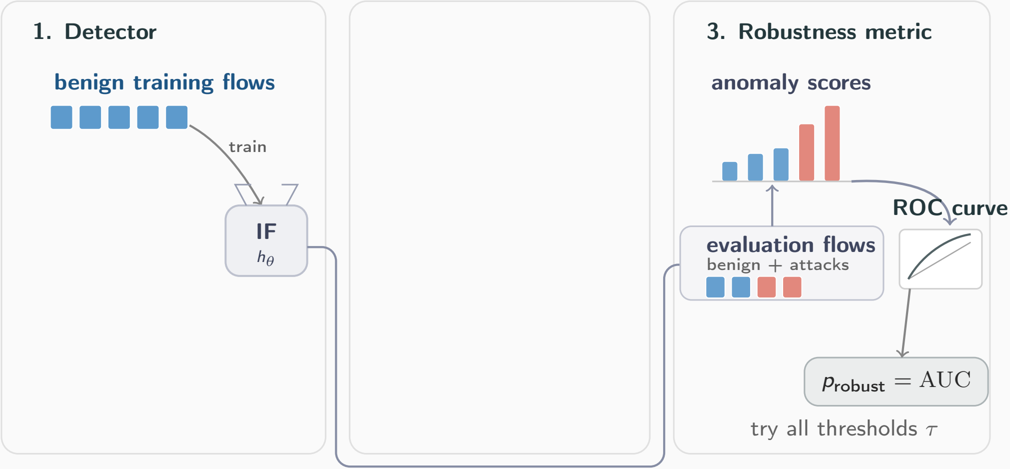

$\Longrightarrow$ ROC–AUC: try all alert thresholds; area under the curve = p_{robust}.
Higher means better attack/benign separation.

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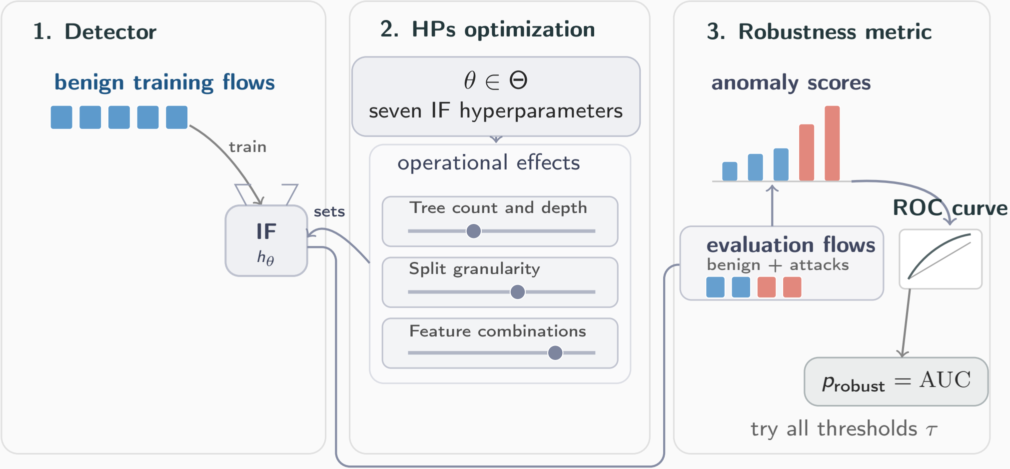
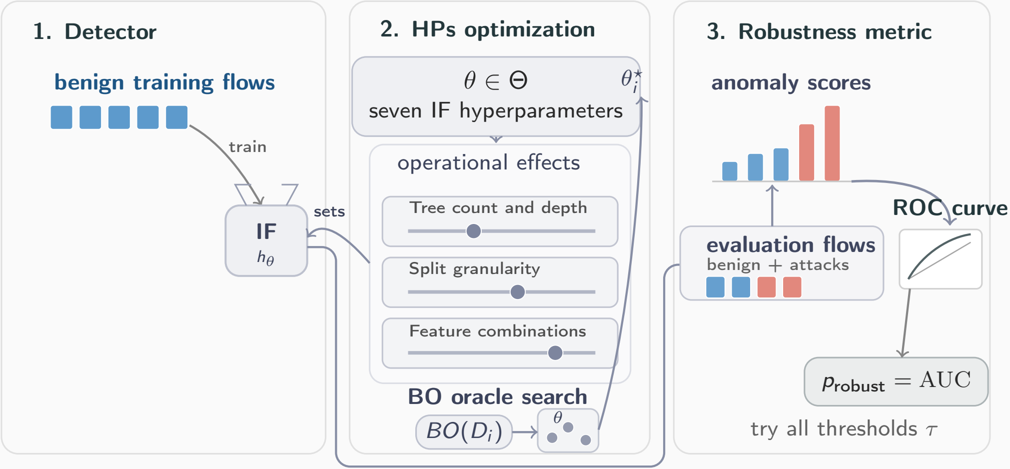


$\Longrightarrow$ θ groups seven IF hyperparameters by their operational effect.

Experimental Setup: IF, BO, and Robustness Metric

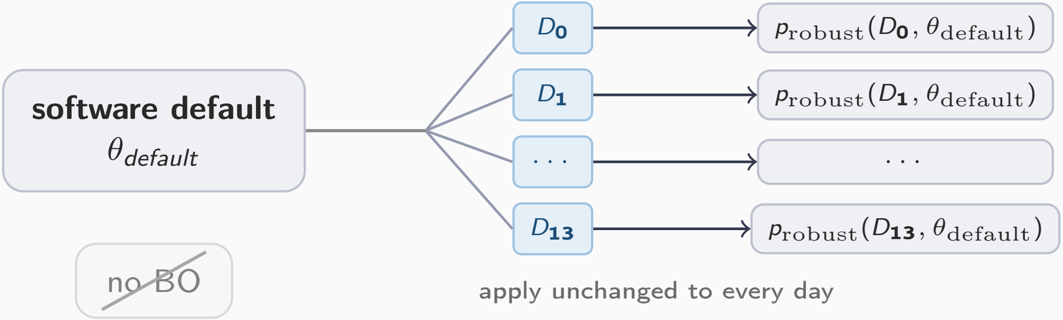


$\Longrightarrow$ BO gives the oracle reference: $\theta_i^\star = BO(D_i)$.

Research Question 1

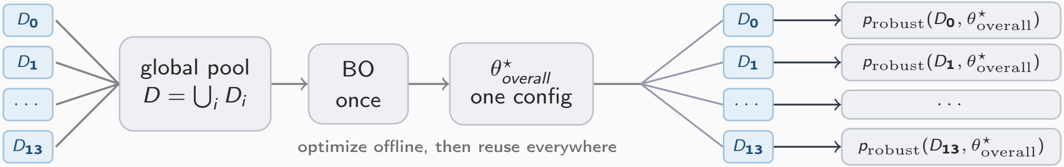
Can cheaper strategies approach an oracle-optimized reference without full reconfiguration?

Experimental Setup: Alternative 1 – Default



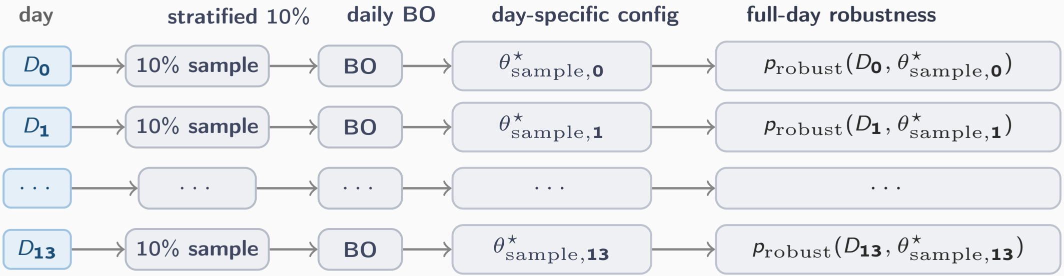
$\Longrightarrow$ Zero tuning cost; tests whether the library default is robust enough across changing network contexts.

Experimental Setup: Alternative 2 – Single Overall



$\Longrightarrow$ Optimize once on heterogeneous historical traffic; amortize BO effort into one reusable configuration.

Experimental Setup: Alternative 3 – 10% Sample BO

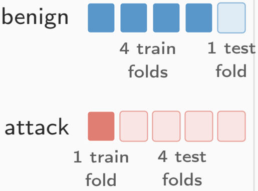


preserve label proportions while reducing the daily BO budget

$\Longrightarrow$ Keep day-specific adaptation, but make BO cheaper by tuning on a stratified 10% subset.

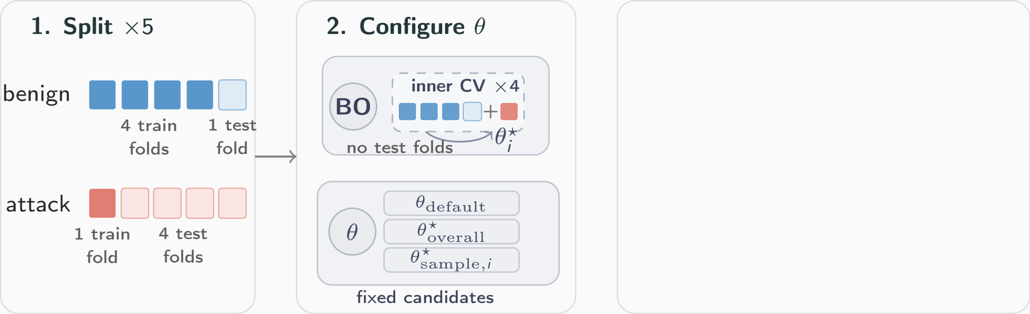
Experimental Setup: Nested Cross-Validation

1. Split $\times 5$



$\Longrightarrow$ Why nested? Choose θ using only training folds, avoiding test leakage.

Experimental Setup: Nested Cross-Validation



$\Longrightarrow$ Configure θ: BO validates on inner benign folds + one attack training fold; reuse skips BO.

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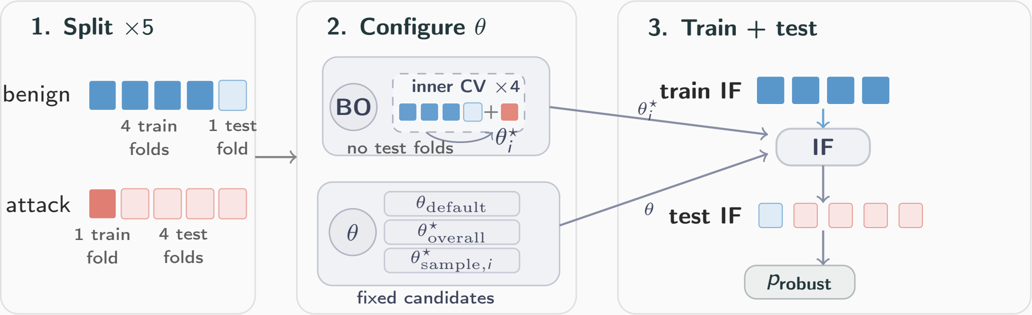

$\Longrightarrow$ Evaluation: IF trains on benign folds, tests on held-out folds, and gives one p_{robust}.

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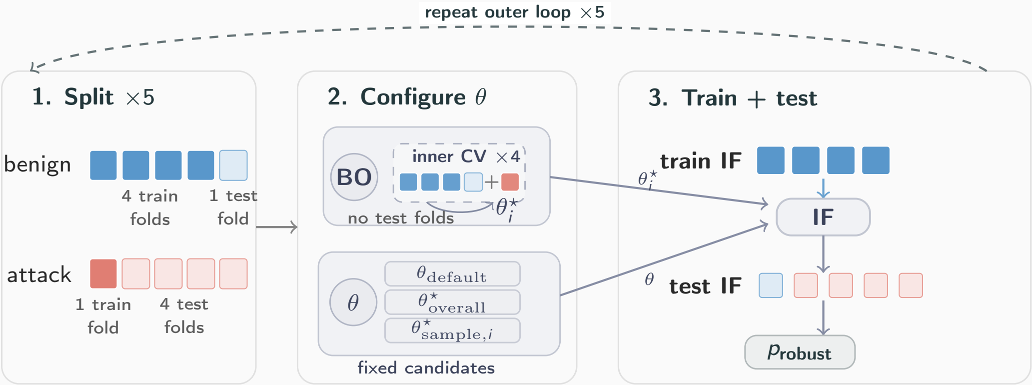

$\Longrightarrow$ Repeat the outer loop over the five folds before averaging.

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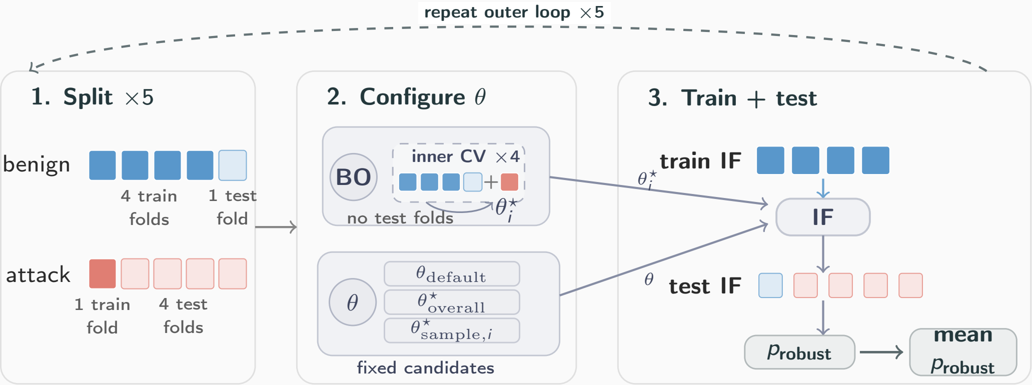
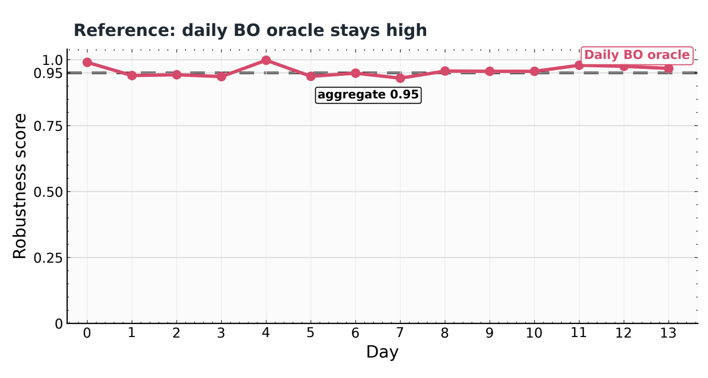


$\Longrightarrow$ The reported score is mean p_{robust}.

Results

Main Results

- The daily BO oracle remains consistently high.



Daily robustness across configuration strategies

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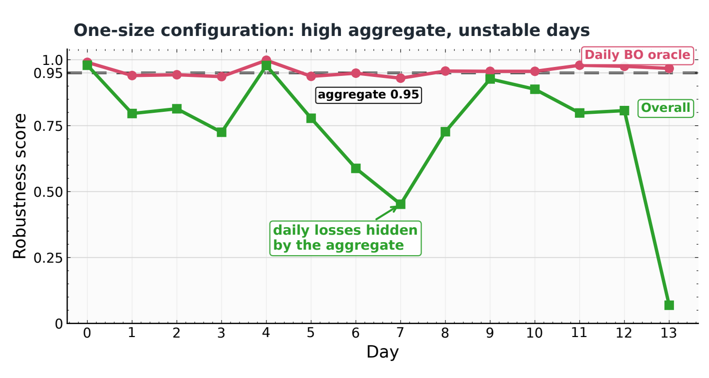

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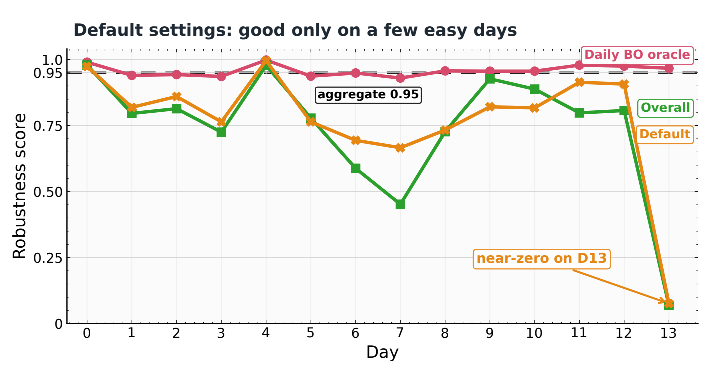

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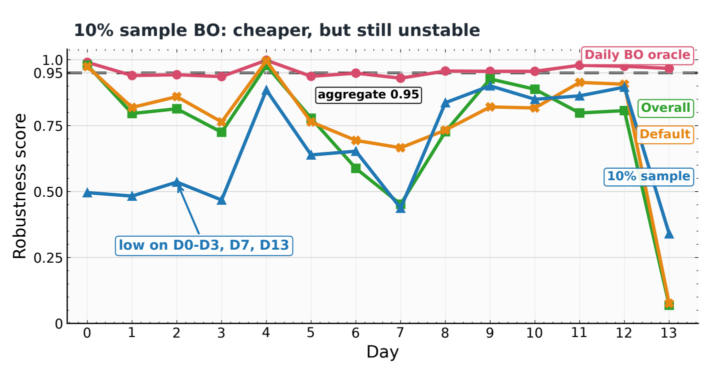

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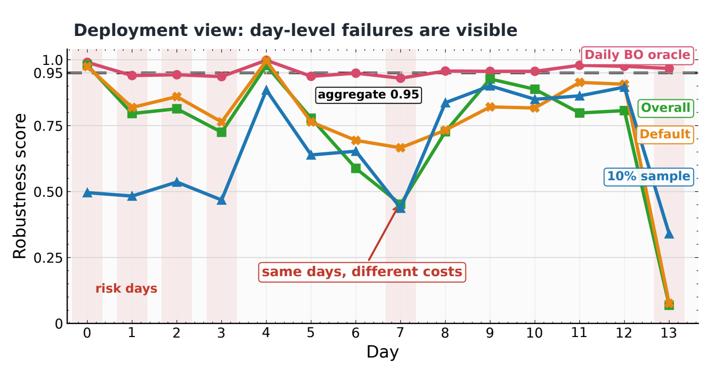
- Day-level evaluation exposes deployment risk.



Daily robustness across configuration strategies

$\Longrightarrow$ Low-budget configurations do not reliably approximate the daily BO oracle.

Research Question 2

If θ is reused,
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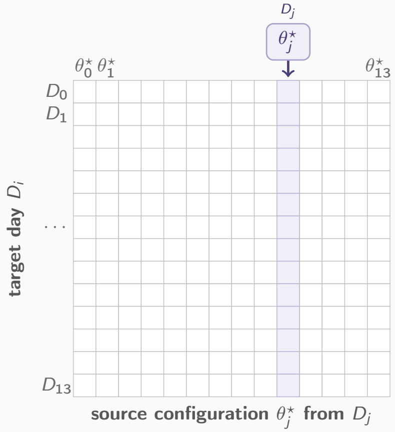
compared with reconfiguring
for the current network context?

Day-to-Day Transferability: Method

- BO gives one oracle configuration per day; pick a source D_j with $\theta_j^\star$.



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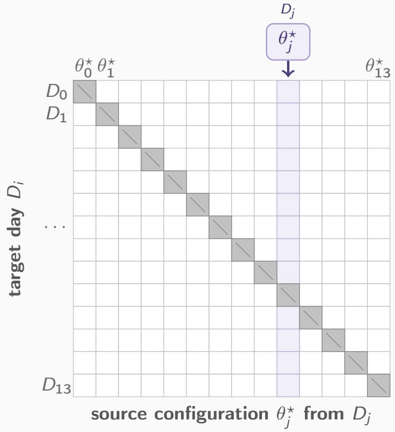

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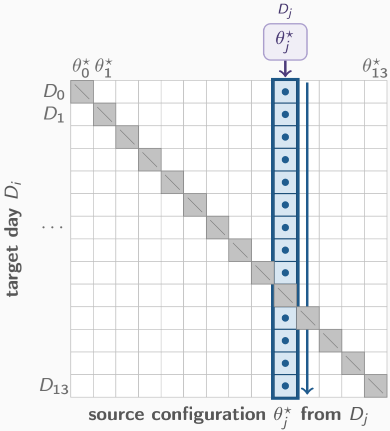

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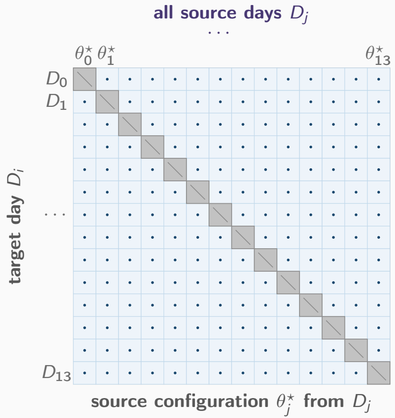

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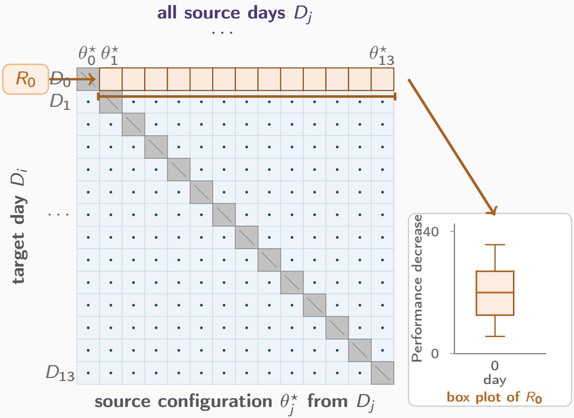
- Example: R_0 collects all transfer losses for target day D_0 and becomes one box plot.

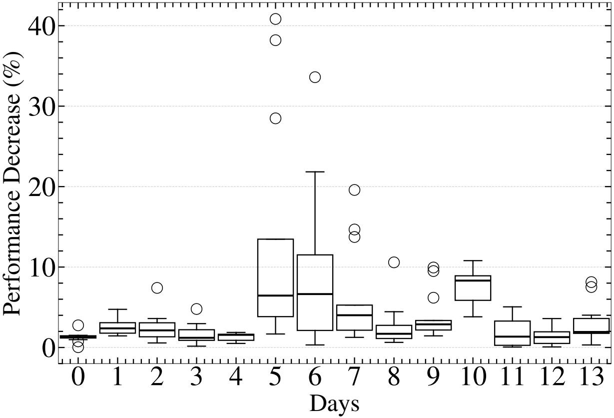
$\Longrightarrow$ Lower row losses indicate safer configuration reuse.



Day-to-Day Transferability: Results

- Each box summarizes one target day $R_i = \{\Delta(i,j) : j \neq i\}$.
- Low medians indicate small transfer loss for most source configurations.
- Wide boxes and outliers show sensitivity to the chosen source day.
- Days 5–7 and 10 are the riskiest; losses can approach 40%.

$\Longrightarrow$ Transferability is day-dependent, not universal.



Transfer loss distributions by target day

Cross-Year Attack Transfer: Method

- Select one of the 18 attack families common to IDS2017 and IDS2018.

common attacks $\mathcal{A}$: same family, different year

a_0 a_1 $\cdots$ a_k $\cdots$ a_{17}

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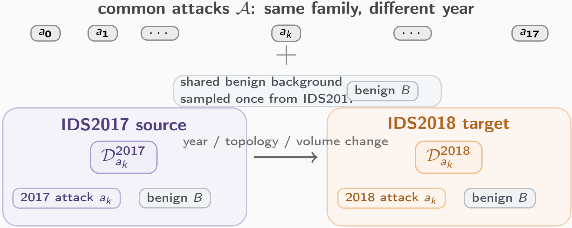

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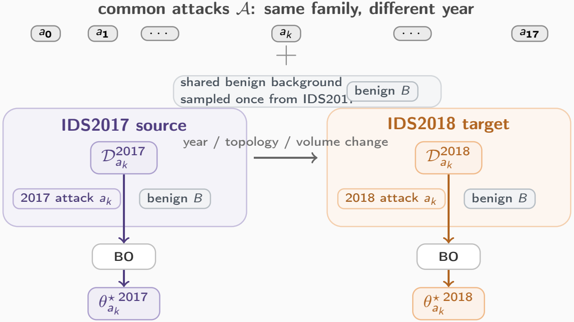

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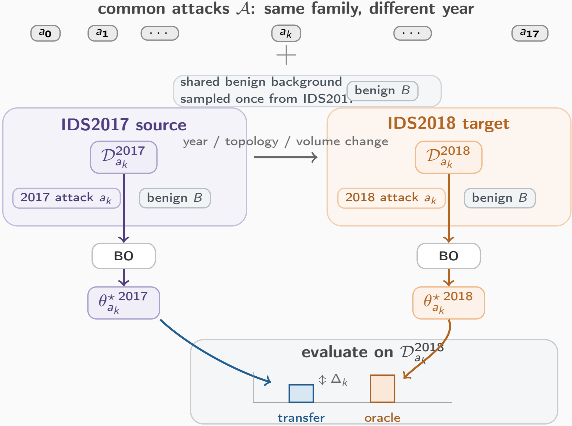
- BO gives $\theta^{\star\,2017}_{a_k}$ and the IDS2018 oracle $\theta^{\star\,2018}_{a_k}$.
- Evaluate both configurations on $\mathcal{D}^{2018}_{a_k}$ and measure the gap.

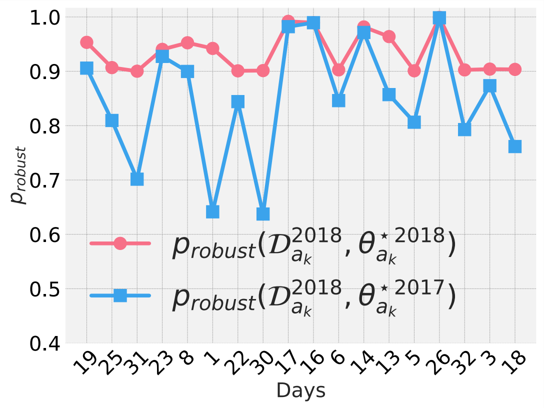
$\Longrightarrow$ Same attack does not imply same robust configuration.



Cross-Year Attack Transfer: Results

- Each attack family compares IDS2017-tuned transfer against the IDS2018 oracle.
- Attacks 23, 17, 16, 14, and 26 transfer close to the oracle.
- Attacks 31, 1, and 30 show much larger cross-year gaps.
- Same attack family is therefore not enough to guarantee robust reuse.

$\Longrightarrow$ Cross-year context changes can still require reconfiguration.



IDS2017 configuration transferred to IDS2018 by attack family

Conclusion

Takeaway

- Retraining alone does not guarantee robustness when hyperparameters stay fixed.
- Configuration reuse under context changes can remove roughly one third of discrimination power.
- AD IDS need automated reconfiguration, not a one-time setup.

Next direction

Trigger reconfiguration only when a lightweight change detector reports enough traffic change.

$\Longrightarrow$ Treat the configuration as an operational asset to monitor.

Acknowledgments

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The opinions expressed are those of the authors and do not necessarily reflect the views of the French government.

Questions?